



Numerical modelling of a sorption-enhanced methanation system

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ARTICLE INFO

Keywords:

Synthetic natural gas
Sorption enhanced methanation
Mathematical model
Bifunctional catalyst
Packed bed reactor
13X zeolite

ABSTRACT

Synthetic natural gas production from CO₂ and green hydrogen provides a promising route to renewable energy storage. Reactants maximum conversion degree, limited by CO₂ methanation strong exothermicity, can be enhanced by in situ water removal. In this work, reaction/adsorption step of a sorption-enhanced methanation process was modelled using a two-dimensional, heterogeneous, and dynamical model of an externally cooled fixed bed reactor. A bifunctional pellet (Ni on 13X zeolite) was considered. Internal/external catalyst mass and heat transfer resistances were assessed. Effect of variations in gas space velocity (GHSV) and operating pressure on produced methane purity and effective operational time length have been evaluated through a sensitivity analysis. Results show that a pure and dry methane flow was produced until a reactant's breakthrough occurs. At any given operating pressure, there is a non-linear negative correlation between GHSV and breakthrough times. Conversely, for any given GHSV, an increase in operating pressure increases breakthrough time.

1. Introduction

According to the Intergovernmental Panel on Climate Change (IPCC (IPCC, 2022)), the greenhouse effect is the result of the worldwide increase in energy utilization and consequential increase in the emission of greenhouse gases (GHG). Over the past decades, scientific research has made great efforts not only to find new technologies to reduce GHG emissions, but also to simultaneously meet the continuously increasing energy demand through a readily available, carbon-neutral source of energy (e.g., (Chen et al., 2022; Wang et al., 2021)). In this framework, renewable energy sources exploitation made significant progress. Nevertheless, most of them continue to suffer of low efficiency, high costs, and technological barriers, mainly related to their intermittent nature. Consequently, large scale energy storage is mandatory to make renewable energy a viable option (Lindley, 2010; Lund, 2018). Besides a wider exploitation of renewable energies, carbon capture and utilization/storage (CCUS) technologies can contribute to CO₂ emissions reduction, allowing the utilization of fossil fuels in a “greener” way (Baena-Moreno et al., 2019; Nocito and Dibenedetto, 2020). Power-to-Methane (PtM) technologies combine renewable energy storage and CCUS (Ghaib and Ben-Fares, 2018; Tregambi et al., 2022), allowing direct conversion of a captured CO₂ stream and the H₂ produced by a renewable source, mainly by using the excess of electrical energy from peaks of production (Leung et al., 2014), to methane through the Sabatier reaction. This latter can be readily injected in the existing

natural gas grid or directly used in an endothermic engine.

Carbon dioxide methanation is an exothermic reaction with significant molar reduction, therefore thermodynamically favoured at low temperature and high pressure. The great challenge in catalytic methanation is to remove the significant heat produced by the Sabatier reaction to both overcome conversion limitation caused by the thermodynamic equilibrium and avoid catalyst sintering (Rönsch et al., 2016). Commercial methanation reactors usually operate within 200–550 °C. The lower temperature limits mainly rely on the activation temperature of the adopted catalyst, while the maximum is dictated by the need to avoid catalyst sintering and carbon formation (Götz et al., 2016; Rönsch et al., 2016). Several kind of reactors have been investigated in literature to limit the temperature increase induced by the exothermic Sabatier reaction (Kopyscinski et al., 2010), ranging from series of packed beds with inter-cooling and water condensation and/or product recycle to fluidized bed reactors and microreactors. Fluidized beds allow almost isothermal operation thanks to their high heat transfer coefficients (Hervy et al., 2021), but high mechanical stresses may enhance catalyst attrition and elutriation which represent a relevant drawback (Rönsch et al., 2016). Therefore, the use of a series of intercooled adiabatic packed-bed reactors with gas recirculation is the most frequent considered solution in literature (Bareschino et al., 2021; Kopyscinski et al., 2010). To increase the conversion, water can be removed by cooling and condensation between consecutive reactors (Giglio et al., 2021). Temperature control in the methanation process can be easily achieved also in micro-structured reactors, consisting of

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<https://doi.org/10.1016/j.ces.2023.118876>

Received 15 March 2023; Received in revised form 28 April 2023; Accepted 10 May 2023

Available online 13 May 2023

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Nomenclature			
A	adsorption potential, $\text{kJ}\cdot\text{kg}^{-1}$	V_p	catalyst particle volume, m^3
a_v	solid specific surface area per unit volume, $\text{m}^2\cdot\text{m}^{-3}$	v	Gas velocity, $\text{m}\cdot\text{s}^{-1}$
c_i	concentration of species i, $\text{mol}\cdot\text{m}^{-3}$	W	adsorption volume, $\text{m}^3\cdot\text{kg}^{-1}$
c_p	heat capacity, $\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$	W_0	maximum adsorption volume, $\text{m}^3\cdot\text{kg}^{-1}$
d_p	particle diameter, m	X_{CO_2}	CO_2 conversion, -
d_t	reactor diameter, m	y	gas molar fraction, -
$D_{\text{ar},i}$	effective radial dispersion coefficient of species i, $\text{m}^2\cdot\text{s}^{-1}$	z	reactor axial coordinate, m
$D_{\text{ax},i}$	effective axial dispersion coefficient of species i, $\text{m}^2\cdot\text{s}^{-1}$	Greek letters	
$D_{\text{eff},i}$	effective diffusion of species i in solid, $\text{m}^2\cdot\text{s}^{-1}$	$\Delta H^\circ_{\text{ads}}$	adsorption enthalpy, $\text{kJ}\cdot\text{mol}^{-1}$
E_{ads}	characteristic adsorption energy, $\text{kJ}\cdot\text{kg}^{-1}$	$\Delta H^\circ_{\text{m}}$	reaction enthalpy, $\text{J}\cdot\text{mol}^{-1}$
F	molar flow rate, $\text{mol}\cdot\text{s}^{-1}$	ε_b	bed porosity, -
f_{ads}	volumetric fraction of sorbent, -	ε_p	solid porosity, -
f_{cat}	volumetric fraction of catalyst, -	η	effectiveness factor of the catalyst, -
GHSV	gas hourly space velocity, $\text{m}^3\cdot\text{kg}_{\text{ads}}^{-1}\cdot\text{h}^{-1}$	λ_{ax}	effective axial heat conductivity, $\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$
h_f	gas–solid heat transfer coefficient, $\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$	λ_{ar}	effective radial heat conductivity, $\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$
h_w	wall heat transfer coefficient, $\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$	λ_s	catalyst thermal conductivity, $\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$
$k_{\text{m},i}$	gas–solid mass transfer coefficient for i-specie, $\text{m}\cdot\text{s}^{-1}$	μ	viscosity, $\text{Pa}\cdot\text{s}$
K_{LDF}	linear driving force coefficient, s^{-1}	ν_i	stoichiometric coefficient
L	reactor length, m	ρ	bulk density, $\text{kg}\cdot\text{m}^{-3}$
MW_i	molar weight of species i, $\text{kg}\cdot\text{mol}^{-1}$	τ	tortuosity, -
Ns	number of species, -	Subscripts, superscripts, and abbreviations	
N_{WP}	Weisz-Prater number, -	0	initial value
P	pressure, bar	ads	adsorbent
P_i	partial pressure, bar	bt	breakthrough
P_{sat}	saturation vapor pressure of adsorbate, Pa	cat	catalyst
Q	water load, $\text{mol}\cdot\text{kg}^{-1}$	cool	cooling
q^*	equilibrium adsorption capacity, $\text{mol}\cdot\text{kg}^{-1}$	CCU	Carbon Capture and Utilization
R_{gas}	gas constant, $\text{J}\cdot\text{mol}^{-1}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$	g	gas phase
R_{m}	methanation reaction rate, $\text{mol}\cdot\text{kg}_{\text{cat}}^{-1}\cdot\text{s}^{-1}$	i	i-species index (i = CH_4 , H_2O , H_2 , CO_2)
r	reactor radial coordinate, m	in	inlet conditions
r_c	pellet coordinate, m	SEM	sorption-enhanced methanation
r_p	pellet radius, m	SNG	Synthetic natural gas
T	temperature, K	STP	standard temperature and pressure, 0° ad 1 bar
t	time, s	s	solid phase
$\dot{V}$	volumetric flow rate, $\text{m}^3\cdot\text{h}^{-1}$		

several micro tubes filled with a catalyst and immersed in a coolant fluid (Brachi et al., 2023; Kreitz et al., 2019). Due to the stoichiometry of the reaction, 5 mol of reactants provide 1 mol of methane on a dry basis. Therefore, even a 99% conversion of CO_2 would only ensure a 95% mole fraction of methane (on a dry basis). This could not be enough to match the regulatory limits for direct grid injection of the produced synthetic natural gas (SNG), as the residual amount of unreacted CO_2 and H_2 is strictly limited by specific country regulations (Erdener et al., 2023). Consequently, the SNG needs to undergo a cleaning process downstream the methanation reactor before its grid injection and this considerably increases the total costs of the whole process (Gorre et al., 2019).

A promising route to overcome equilibrium limitations in the Sabatier reaction is represented by sorption-enhanced methanation (SEM), where selective removal of produced water occurs simultaneously with the methanation reaction. Water removal can be obtained by a reactor integrated with a water vapor permselective membrane (Schlereth and Hinrichsen, 2014). For this process to be efficient, the pressure must be above 10 bar (Schlereth and Hinrichsen, 2014). More recently, several authors investigated the use of zeolites for water adsorption showing that quite long running times can be reached, even at temperature up to 350°C . In details, Walspurger et al. (Walspurger et al., 2014) analysed the performance of a physical mixtures of a sorbent (4A zeolite) and a Ni-based catalyst, showing that 100% conversion of H_2 to methane can be reached even at atmospheric pressure. Wei et al. (Wei et al., 2021b, Wei et al., 2021a) used different zeolites (13X, 5A and

L) to support a Ni catalyst in a bifunctional material. In both cases, a complete CO_2 conversion was reported by the authors, even for temperature of 320°C , also reporting also that gas hourly space velocity (GHSV) does not significantly affect the water adsorption capacity. Delmelle et al. (Delmelle et al., 2016) tested a 5A and a 13X zeolite doped with 5 %wt Ni at atmospheric pressure and operating temperature of 300°C , showing that a complete CO_2 conversion can be attained. More recently, Barcala et al. (Cañada-Barcala et al., 2023) showed that CO_2 conversion degree and methane selectivity greater than 99% can be attained at atmospheric pressure and even for temperature of 350°C using a commercial Ni-based catalyst mixed with 3A zeolite. Pressure values greater than 1 bar were considered too (Agirre et al., 2021). To increase the catalytic effect, especially at low temperatures, noble metals such as ruthenium and rhodium were considered for CO_2 methanation (Rönsch et al., 2016). Gomez et al. (Gómez et al., 2022) studied SEM using a 5A zeolite mixed to a Rh-based catalyst, feeding a stoichiometric mixture of H_2/CO and H_2/CO_2 . Syngas and CO_2 conversions close to 100% were obtained even at atmospheric pressure. Kiefer et al. (Kiefer et al., 2022) recently proposed a mathematical model for pellet bed reactors by using an integral mathematical model in order to address the effect of the main parameters of SEM. A 13X zeolite impregnated by Ni was used as bifunctional material. As the authors reported, the thermal design of the reactor is of utmost importance and more sophisticated models are essential for an accurate description of the process. The main problem when using a zeolite for in-situ water

removal is the high temperature that can be achieved during methanation. Temperatures higher than 400 °C are detrimental for the process, as the ability to adsorb water strongly decreases above this latter value (Wei et al., 2021a). However, commercial methane production processes take place in a cascade of reactors, with the last one receiving very low concentrations of reactants, hence the heat produced by methanation is significantly lower than that produced in the first reactors of the series (Bareschino et al., 2021). Consequently, the temperatures achieved in the last reactor do not exceed 400 °C, allowing the use of zeolites for SEM (Walspurger et al., 2014).

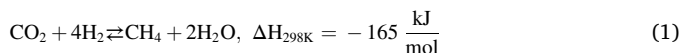
Intensification through a sorption enhanced reaction with a zeolite acting as adsorbent was applied in hydrogenation of CO₂ to methanol (Maksimov et al., 2021) and to dimethyl-ether (DME) (An et al., 2008; van Kampen et al., 2021). Guffanti et al. (Guffanti et al., 2021b) developed a 2D heterogeneous dynamic model of an externally cooled packed bed reactor to investigate the effect of main operating and design parameters such as reactor dimensions and adsorbent-to-catalyst weight ratio. The reactor was considered filled by a mechanical mixture of a 3A zeolite and a Cu/Zn/Al₂O₃ catalyst, and the distribution of the two active phases in hybrid or core@shell pellets is addressed (Guffanti et al., 2021a). Sorption enhanced DME synthesis is slightly affected by CO/CO₂ feed ratio, and an increase of CO₂ content in the feed provides larger advantages with respect to conventional DME synthesis (Guffanti et al., 2021b). Internal diffusion in catalyst particles have a strong influence on effective reaction rates (Guffanti et al., 2021a). By means of a pseudo-homogenous reactor model, Maksimov et al. (Maksimov et al., 2022, 2021) addressed the efficiency of methanol synthesis via sorption enhanced reaction in a fixed bed reactor filled with a mechanical mixture of zeolite 3A and a Cu/Zn/Al₂O₃ catalyst. It was found that methanol production rate strongly depends on GHSV. In all the cases, it is pointed out that, due to the loss of adsorption capacity at increasing temperatures, heat management is a crucial point in sorption enhanced process. This problem is even more severe for methane synthesis as the heat released by the methanation reaction is much greater than for DME and methanol synthesis.

To the best of the Authors' knowledge, there are no studies in literature discussing reactor modelling for the SEM process using zeolites. Therefore, there is a specific need for both numerical and analytical tools that may support the experimental activity, simplify the scale-up process, and reduce the high costs of the experiments. In this paper, a detailed mathematical model for 13X-based SEM has been described. The process under investigation is carried out in a packed bed reactor with external cooling. The mathematical model is 2D, heterogeneous, and dynamic, since only the time interval during which the solid is

2. Mathematical model

A single tube of a multi-tubular cooled reactor for SEM was described using a 2D, dynamic, and heterogeneous model. The proposed model describes axial and radial variations in the temperature and concentrations of all chemical species and accounts for inter- and intra-phase gradients. The mathematical model is dynamic because the SEM process is inherently non-stationary. In the present work, the mathematical model focuses on the adsorption/reaction cycle of a SEM process. The step concerning water desorption and, thus, solid regeneration has not been described.

The CO₂ methanation reaction (1), i.e., the Sabatier reaction, is a highly exothermic equilibrium reaction, whose products are CH₄ and H₂O.



A possible formation of CO may occur by reverse water gas shift reaction which, however, is appreciable only at temperatures above 450 °C (Borgschulte et al., 2013). For the case studied in the present work, the fed mixtures have a high CH₄ content (80 %_v at least), so the maximum achievable temperatures are below 450 °C and CO formation can be safely neglected.

Another problem that can occur at high temperatures and low pressures is the formation of solid carbon, which leads to catalyst deactivation and methane yield reduction (Götz et al., 2016). Although, as previously discussed, the maximum temperature considered never exceeds 450 °C, in-situ water removal could theoretically result in carbon formation even at this operating condition. However, in current literature, several authors experimentally investigated SEM and in no case witnessed the presence of solid carbon (Walspurger et al., 2014; Wei et al., 2021a). Therefore, the formation of solid carbon was neglected in this work, and the only species considered were CH₄, CO₂, H₂, and H₂O.

2.1. Gas and solid phase mass balances

The mass balances for gas and solid phase are expressed on molar basis and consist in a set of partial differential equations embodying mass transfer in axial and radial direction for gas phase and along pellet radius for solid phase, mass exchange between the gas and solid phase, and reaction/adsorption in the catalyst. In details, the 2D-dynamical mass balances for *i*th chemical species (CH₄, CO₂, H₂, H₂O) in the bulk gas phase considered axial and radial dispersion, axial convection, and

$$\varepsilon_b \frac{\partial C_{i,g}}{\partial t} = \varepsilon_b D_{a,r,i} \left(\frac{\partial^2 C_{i,g}}{\partial r^2} + \frac{1}{r} \frac{\partial C_{i,g}}{\partial r} \right) + \varepsilon_b D_{a,z,i} \frac{\partial^2 C_{i,g}}{\partial z^2} - \frac{\partial(vC_{i,g})}{\partial z} - k_{m,i} a_v \left(C_{i,g} - C_{i,s} \Big|_{r_c=d_{p/2}} \right) \quad (2)$$

capable of adsorbing water is of interest. The reactor filling considered is a bifunctional material consisting in a 13X zeolite to promote in-situ water removal loaded with Ni acting as a catalyst. According to Mills and Steffgen (Mills and Steffgen, 1974), Ni catalysts exhibit high activity and selectivity towards the CO₂ methanation and, due to comparatively low price, are the most commonly active metal industrially used. 13X zeolite represents an excellent candidate for SEM (Kiefer et al., 2022; Wei et al., 2021a, Wei et al., 2021b), showing a high-water uptake capacity and, as reported by Delmelle et al. (Delmelle et al., 2016), allowing for longer operation times with respect to other zeolite-supported catalysts.

mass transfer between gas and solid phase. They were expressed as:

The dynamical overall mass balance equation only considered convection and mass transfer between the gas and solid phase:

$$\varepsilon_b \frac{\partial C_{\text{tot},g}}{\partial t} = - \frac{\partial(vC_{\text{tot},g})}{\partial z} - \sum_{i=1}^{N_s} k_{m,i} a_v \left(C_{i,g} - C_{i,s} \Big|_{r_c=d_{p/2}} \right) \quad (3)$$

In Eqs. (2)-(3), the solid specific surface area was calculated as:

$$a_v = \frac{6}{d_p} (1 - \varepsilon_b) \quad (4)$$

while for the bed porosity, the Dixon (Dixon, 1988) correlation for a

spheric particle was adopted:

$$\varepsilon_b = 0.4 + 0.05 \left(\frac{d_p}{d_t} \right) + 0.412 \left(\frac{d_p}{d_t} \right)^2 \quad (5)$$

The momentum balance is governed by the friction factor, thus the pressure drop along the bed axis can be described by the Ergun equation. Nonetheless, preliminary analyses revealed that, due to the low gas velocities considered, the pressure drop is well below 5% and can thus be safely neglected.

Dynamical mass balance in the solid phase for CH₄, CO₂, and H₂ considered the simultaneous diffusion and reaction:

$$\varepsilon_p \frac{\partial C_{i,s}}{\partial t} = \varepsilon_p D_{eff,i} \left(\frac{\partial^2 C_{i,s}}{\partial r_c^2} + \frac{1}{r_c} \frac{\partial C_{i,s}}{\partial r_c} \right) + \rho_{cat} f_{cat} \nu_i R_m \quad (6)$$

while mass balance in the solid phase for H₂O accounted for adsorption too:

$$\varepsilon_p \frac{\partial C_{H_2O,s}}{\partial t} = \varepsilon_p D_{eff,i} \left(\frac{\partial^2 C_{H_2O,s}}{\partial r_c^2} + \frac{1}{r_c} \frac{\partial C_{H_2O,s}}{\partial r_c} \right) + \rho_{cat} \nu_{H_2O} R_m - \rho_{ads} f_{ads} \frac{\partial q}{\partial t} \quad (7)$$

At the typical temperatures at which CO₂ methanation takes place, H₂O and CO₂ co-adsorption is negligible (Iliuta et al., 2011), thus just water adsorption was considered.

In Eqs. (6)-(7), the reaction and adsorption terms are weighed by the volumetric fraction per unit volume f_{cat} and f_{ads} , respectively. Appropriate boundary conditions complete the system of partial differential equations. Specifically, for the gas phase, typical Danckwerts boundary conditions were imposed at the reactor inlet and outlet:

$$C_{i,g,in} = C_{i,g} \Big|_{z=0} - \frac{D_{ax,i}}{v} \frac{\partial C_{i,g}}{\partial z} \Big|_{z=0} \quad (8)$$

$$\frac{\partial C_{i,g}}{\partial z} \Big|_{z=L} = 0 \quad (9)$$

Axial symmetry and impermeability completed the boundary conditions:

$$\frac{\partial C_{i,g}}{\partial r} \Big|_{r=0} = \frac{\partial C_{i,g}}{\partial r} \Big|_{r=\frac{d_p}{2}} = 0 \quad (10)$$

The boundary conditions for the overall mass balance were:

$$(v C_{tot,g})_{z=0} = (v C_{tot,g})_{in} \quad (11)$$

Symmetry and flux continuity represented the boundary conditions for the solid phase:

$$\frac{\partial C_{i,s}}{\partial r_c} \Big|_{r_c=0} = 0 \quad (12)$$

$$-D_{eff,i} \frac{\partial C_{i,s}}{\partial r_c} \Big|_{r_c=d_p/2} = k_{m,i} \left(C_{i,g} - C_{s,i} \Big|_{r_c=d_p/2} \right) \quad (13)$$

2.2. Gas ad solid phase enthalpy balance

The enthalpy balances for gas and solid phase consist in a set of partial differential equations embodying heat transfer in axial and radial direction for gas phase, heat exchange between gas and solid phases, and reaction/adsorption heat generation in the catalyst. In details, the 2D-dynamical energy balance considered axial and radial thermal dispersion, axial convection, and heat transfer between gas and solid phase:

$$\varepsilon_b \rho_g c_{p,g} \frac{\partial T_g}{\partial t} = \lambda_{ar} \left(\frac{\partial^2 T_g}{\partial r^2} + \frac{1}{r} \frac{\partial T_g}{\partial r} \right) + \lambda_{ax} \frac{\partial^2 T_g}{\partial z^2} - \rho_g c_{p,g} v \frac{\partial T_g}{\partial z} + h_f a_v (T_s - T_g) \quad (14)$$

Due to the high thermal conductivity of the solid, intraparticle temperature gradients can be neglected (Guffanti et al., 2021b) and the

energy balance for the solid phase was written as:

$$(\rho_{cat} f_{cat} + \rho_{ads} f_{ads}) c_{p,s} \frac{\partial T_s}{\partial t} = h_f a_v (T_s - T_g) + \rho_{cat} f_{cat} (-\Delta H_m) R_m + \rho_{ads} f_{ads} (-\Delta H_{ads}) \frac{\partial q}{\partial t} \quad (15)$$

To complete the gas and solid enthalpy balances, classic Danckwerts boundary conditions were imposed at the reactor inlet and outlet:

$$T_{g,in} = T_g \Big|_{z=0} - \frac{\lambda_{ax}}{\rho_g c_{p,g} v} \frac{\partial T_g}{\partial z} \Big|_{z=0} \quad (16)$$

$$\frac{\partial T_g}{\partial z} \Big|_{z=L} = 0 \quad (17)$$

Symmetry condition and heat flux continuity at the wall were imposed for radial boundaries:

$$\frac{\partial T_g}{\partial r} \Big|_{r=0} = 0 \quad (18)$$

$$\frac{\partial T_g}{\partial r} \Big|_{r=d_p/2} = \frac{h_w (T_{cool} - T_g)}{\lambda_{ar}} \quad (19)$$

Finally, being the study focused only on the adsorption/reaction step, the initial condition refers to solid with no water adsorbed, while spatial uniform temperature and concentration, equal to the inlet one, are assumed as initial conditions for heat (Eqs. (14)-(15)) and mass (Eqs. (2)-(3) and (6)-(7)) balances, respectively.

The Shomate equation and data from the NIST Chemistry WebBook (IUPAC, 2007) were used to calculate heat capacities and enthalpy of formation of all the gaseous species. The temperature dependence of mass and heat transport coefficients, diffusivities, and gas properties were considered through the state-of-the-art correlations (see [Supplementary materials - Section S1-S3](#)).

2.3. Reaction kinetic and adsorption isotherm

The solid phase consists of a bifunctional material on which simultaneously CO₂ methanation is catalysed and in-situ water removal occurs. Kinetics of water adsorption on the 13X zeolite is described by a linear driving force (LDF) model (Mette et al., 2014):

$$\frac{\partial q}{\partial t} = K_{LDF} (q^* - q) \quad (20)$$

where the equilibrium adsorption capacity, q^* , is described by the Dubinin Ashtakov model (Mette et al., 2014):

$$W = W_0 \exp \left[- \left(\frac{A}{E_{ads}} \right)^n \right] \quad (21)$$

$$A = \frac{R_{gas}}{MW_{H_2O}} T \ln \left(\frac{P_{sat}}{P_{H_2O}} \right) \quad (22)$$

The volumetric adsorption capacity W is converted to molar adsorption capacity through the density of the adsorbed phase $\rho(T)$:

$$q^* = \frac{\rho(T)}{MW_{H_2O}} W(T, P) \quad (23)$$

Finally, the linear driving force coefficient, accounting for internal

Table 1
Adsorption isotherm parameters for the Dubinin Ashtakov model.

Parameter	Value (Mette et al., 2014)	Value (Kiefer et al., 2022)
W_0 [m ³ •kg ⁻¹]	341•10 ⁻⁶	90.17•10 ⁻⁶
E_{ads} [kJ•kg ⁻¹]	1192.25	1030.9
n	1.55	1.55

mass transfer resistances too, is calculated as (Mette et al., 2014):

$$K_{LDF} = \frac{15}{0.5d_p^2} \frac{D_{M,H_2O} MW_{H_2O} \epsilon_p}{\tau \rho_{ads} R_{gas} T} \frac{1}{\partial q^* / \partial P_{H_2O}} \quad (24)$$

Table 1 reports two set of parameters for the Dubinin Ashtakov model, one proposed by Mette et al. for temperatures up to 250 °C (Mette et al., 2014; Mollo-Varillas et al., 2022) and the other one by Kiefer et al. (Kiefer et al., 2022) for temperatures higher than 300 °C (see Supplementary material S.4 for further details).

In the present work, an average constant value was assumed for the heat released during adsorption of H₂O.

Several CO₂ methanation reaction mechanisms over Ni-based catalyst were proposed in literature (Rönsch et al., 2016). The kinetic model proposed by Koschany et al. (Koschany et al., 2016) for Ni-catalysed CO₂ methanation, which neglects side reactions, was adopted in this work. Moreover, these kinetics were recently used to model a SEM process in (Kiefer et al., 2022). For Readers' convenience, the reaction kinetic is:

$$R_m = \frac{k(P_{H_2} P_{CO_2})^{0.5} \left(1 - \frac{P_{CH_4} P_{H_2O}}{P_{H_2}^4 P_{CO_2} K_{eq}}\right)}{\left(1 + K_{OH} \frac{P_{H_2O}}{P_{H_2}^{0.5}} + K_{H_2} P_{H_2}^{0.5} + K_{mix} P_{CO_2}^{0.5}\right)^2} \quad (25)$$

where the equilibrium constant (K_{eq}) is approximated by the empirical formula (Koschany et al., 2016):

$$K_{eq} = 137T^{-3.994} \exp\left(\frac{158.7 \text{ kJ/mol}}{RT}\right) \quad (26)$$

The mathematical model was solved using the commercial software package COMSOL Multiphysics®. The system domain was discretized by physically controlled mesh with 1000 domain and 420 boundary elements for the bulk phase, while a cubic root sequence of 16 nodes is used to discretize radial axis in the pellet. It was carefully checked that further refinements do not produce any changes in model solutions.

3. Model validation

The proposed reactor model was validated against experimental data available in literature. First, the adsorption isotherm and the methanation kinetics were validated by comparing the H₂O adsorption capacity at breakthrough and the CO₂ conversions predicted by the mathematical model with the experimental data available in the most recent literature. Then, the dynamic feature of the model was validated by comparing the characteristic breakthrough times of the main chemical species.

As discussed in Section 2, a Dubinin-Ashtakov model was used to predict the temperature and pressure dependence of the water equilibrium loads. Both sets of parameters reported in Table 1 were checked in this study, and the H₂O breakthrough capacities predicted by the model were benchmarked against experimental data (Wei et al., 2021a) in isothermal conditions for several operating temperatures. The result of this comparison was summarized in Fig. 1 where, according to Wei et al. (Wei et al., 2021a), the feed gas composition for the numerical simulation was 6% N₂, 10% H₂, 2.5% CO₂, 81.5% CH₄ by volume. By using the values proposed by Mette et al. (Mette et al., 2014), the mathematical model overestimates the solid adsorption capacity, but the difference between the experimental and predicted data decreases as the temperature increases. On the other hand, employing the values recently proposed by Kiefer et al. (Kiefer et al., 2022), the H₂O breakthrough capacity was significantly underestimated (greater than 50%) for all the temperature values. As the temperature change induced by the exothermic CO₂ methanation cannot be neglected in real operation, the reactor works at temperatures, on average, greater than 320 °C. Therefore, in this work the W_0 and E_{ads} values proposed by Mette et al. (Mette et al., 2014) were used to describe the adsorption isotherm. With this set of parameters, the CO₂ conversion attained when H₂O breakthrough capacity was reached was also calculated. Fig. 1(b) reports the CO₂ conversion for non-SEM and SEM obtained by the mathematical model, and the experimental one (Wei et al., 2021a). The equilibrium curve reported in Fig. 1(b) was obtained by minimizing the Gibbs energy of the system (Bareschino et al., 2020). As it can be seen in Fig. 1(b), the CO₂ conversion is slightly overestimated by the model at low temperature values for a non-SEM process. On the other hand, for a SEM process, the

Table 2

Breakthrough time for two operating temperatures. According to Walspurger et al. (Walspurger et al., 2014), the feed gas composition is 6% N₂, 9.9% H₂, 2.5% CO₂, 81.5% CH₄ by volume, at P = 1 bar, GHSV = 2500 $\frac{\text{mL}}{\text{g}_s \text{ h}}$, zeolite-to-catalyst mass ratio = 5:1.

T [°C]	H ₂ O breakthrough [min]		H ₂ breakthrough [min]	
	Experimental (Walspurger et al., 2014)	Predicted	Experimental (Walspurger et al., 2014)	Predicted
300	14	23	10.5	12
350	11.3	14	8.8	9.5

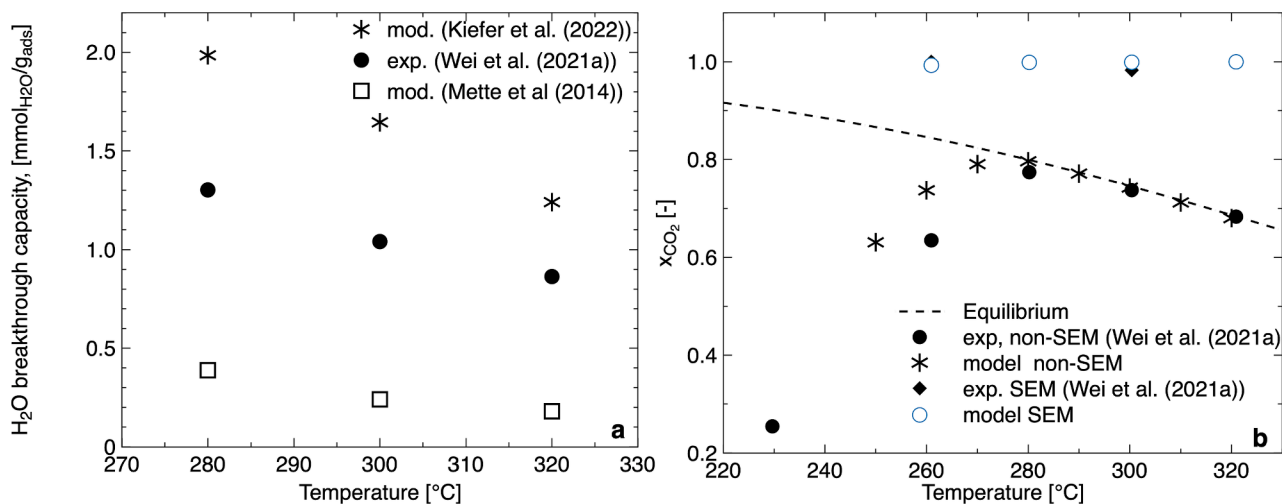


Fig. 1. (a): H₂O breakthrough capacity vs temperature for P = 1 bar, GHSV = 923 $\frac{\text{mL}}{\text{g}_s \text{ h}}$, and Ni-13X with a metal loading of 5 %wt.; Fig. 1(b) CO₂ conversion vs temperature for non-SEM and SEM process for P = 1 bar, GHSV = 714 $\frac{\text{mL}}{\text{g}_s \text{ h}}$, and Ni-13X with a metal loading of 5 %wt. In the legend exp. and mod. stay for experimental and model.

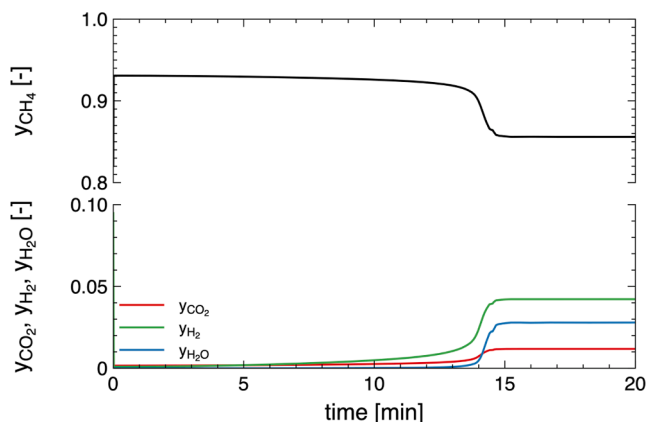


Fig. 2. Gas molar fraction at the reactor outlet at $T = 350\text{ }^{\circ}\text{C}$; feed gas composition, space velocity, and pressure are those reported in Table 2 caption.

conversions predicted by the model perfectly matched the experimental ones.

To the best of the Authors' knowledge, no literature data are available to exactly compare the dynamic response of the proposed mathematical model. Indeed, Gomez *et al.* (Gómez *et al.*, 2022) reported reactor outlet concentrations for a SEM process of CO_2 , but employing Rh as catalyst and a 5A zeolite as adsorber. Wei *et al.* (Wei *et al.*, 2021a, 2021b) reported the outlet compositions using a packed bed reactor but for a bifunctional material in which a 13X zeolite is mixed with Ni/Ce (5/2.5%) catalyst. Walspurger *et al.* (Walspurger *et al.*, 2014) considered a Ni-based catalyst on 4A zeolite. Table 2 shows the comparison between experimental (Walspurger *et al.*, 2014) and predicted breakthrough times of water and hydrogen, defined as the time at which 10% of the maximum concentration is registered at the reactor exit.

Hydrogen breakthrough time was predicted by the model with satisfactory accuracy, with a difference always lower than 1.5 min. For water, a difference of 2.7 min was obtained at $350\text{ }^{\circ}\text{C}$ (i.e., 20%), and of 9 min at $300\text{ }^{\circ}\text{C}$ (40%). These discrepancies can be justified considering that the experimental data refers to a 4A zeolite, while in the present work a 13X zeolite was considered, which, due to its larger pore size, has higher adsorption capacity (Delmelle *et al.*, 2016; Iliuta *et al.*, 2011; Wei *et al.*, 2021a). Moreover, it should be recalled that the mathematical model slightly overestimated the adsorption capacity at $300\text{ }^{\circ}\text{C}$ (see Fig. 1(a)), as previously stated.

Finally, by analysing the time series of reactants and products depicted in Fig. 2, which refers to an isothermal operation, it is possible to find peculiar qualitative behaviours discussed in literature for SEM. Indeed, H_2 and CO_2 breakthroughs always precede the H_2O one, and this is more evident for H_2 (Walspurger *et al.*, 2014; Wei *et al.*, 2021a). Besides, when H_2O breakthrough occurred, H_2 , CO_2 , and CH_4 reached the equilibrium concentrations. Finally, a complete CO_2 conversion (6.5% of N_2 at reactor exit not reported in Fig. 2) was observed at the reactor outlet until H_2 breakthrough was attained ($t \approx 10\text{ min}$) remaining above 98% until water breakthrough occurs ($t \approx 14\text{ min}$).

4. Results and discussion

Section 4.1 discusses the general behaviour of the cooled packed bed in which the SEM occurs. Section 4.2 addresses internal heat/mass transfer limitations, while Section 4.3 assesses and critically discusses the effects of a variation in the main process parameters (namely, gas space velocity and operating pressure) on system performance.

As discussed above, the reactor modelled in this work is the last one of a cascade, so that milder temperature values allowing water adsorption are achieved. Moreover, it was assumed that the stream fed undergoes a preliminary water condensation step. Accordingly, the feed gas composition considered in this study was CH_4 (80%), and H_2 and

Table 3

Reactor parameters, operating condition, and solid properties.

Parameter	Value
ΔH_m°	$-165 \cdot 10^3\text{ J}\cdot\text{mol}^{-1}$
$\Delta H_{\text{ads}}^\circ$	$-45 \cdot 10^3\text{ J}\cdot\text{mol}^{-1}$
$c_{p,s}$	$1100\text{ J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$
λ_s	$0.25\text{ W}\cdot\text{m}\cdot\text{K}^{-1}$
L	2 m
d_t	0.05 m
d_p	2.5 mm
M_{cat}	0.064 kg
M_{ads}	1.22 kg
ρ_{ads}	$1400\text{ kg}\cdot\text{m}^{-3}$
ρ_{cat}	$2440\text{ kg}\cdot\text{m}^{-3}$
T_{in}	$300\text{ }^{\circ}\text{C}$
T_{cool}	T_{in}
T_0	T_{in}
P	1 bar
GHSV	$0.5\text{ m}^3\cdot\text{kg}_{\text{ads}}^{-1}\cdot\text{h}^{-1}$

CO_2 in stoichiometric ratio, 16%_v and 4%_v, respectively. According to Koshany *et al.* (Koshany *et al.*, 2016), for a Ni-based catalyst, $300\text{ }^{\circ}\text{C}$ is a reasonable trade-off for the catalytic activity to take place, therefore both the inlet and the initial reactor temperature were set at $300\text{ }^{\circ}\text{C}$. The temperature of the external coolant was assumed to be constantly equal to the inlet temperature. The catalyst-to-zeolite weight ratio was chosen to be 5%_{wt} (Borgschulte *et al.*, 2013). The other main geometrical and operating parameters used in the present study were reported in Table 3.

For results discussion, GHSV was defined, consistently with SEM-related literature, as the volume flow rate of reactant gases at standard conditions over the solid mass used in the reactor:

$$\text{GHSV} = \frac{\dot{V}_{\text{STP}}}{M_{\text{ads}}} \quad (27)$$

The breakthrough time ($t_{\text{BT},i}$) for the i^{th} -species is defined as the time instant at which the gas composition reaches the 10% of the maximum at the reactor outlet, where the maximum composition is calculated as the equilibrium composition at process temperature and pressure. Finally, CO_2 conversion is calculated as:

$$X_{\text{CO}_2} = \frac{F_{\text{CO}_2,\text{in}} - \bar{F}_{\text{CO}_2,\text{out}}}{F_{\text{CO}_2,\text{in}}} \quad (28)$$

where $\bar{F}_{\text{CO}_2,\text{out}}$ represents the averaged molar flow rate at the outlet section. Being the reactive process described just by CO_2 methanation reaction (Eq.1), the methane yield coincides with CO_2 conversion while the selectivity is 100%.

4.1. SEM: Numerical results

Fig. 3 shows reactor temperature distribution at several times. The catalyst layers closest to the reactor inlet are those that continuously receive a fresh flow of reactants. Therefore, the methanation reaction is faster and higher temperatures are reached here. Moreover, at the beginning ($t = 5\text{ min}$ in Fig. 3), both exothermic methanation and water adsorption occur: in-situ water removal shifts the equilibrium toward the products and a complete reactants conversion occurs ($X_{\text{CO}_2} = 1$), further increasing the heat produced by methanation. Therefore, the highest temperatures ($\approx 400\text{ }^{\circ}\text{C}$) were reached at the beginning of the process and in sections close to the inlet (Fig. 3, $t = 5\text{ min}$).

It is important to underline that the solid has a bifunctional effect. Thus, when in a certain section the maximum adsorption capacity is attained, the solid still acts as catalyst, and the region where the reaction takes place in SEM conditions shifts downstream. Therefore, when the catalyst layers close to the inlet one reach the maximum water adsorption capacity, the methanation reaction still goes on, but no longer under

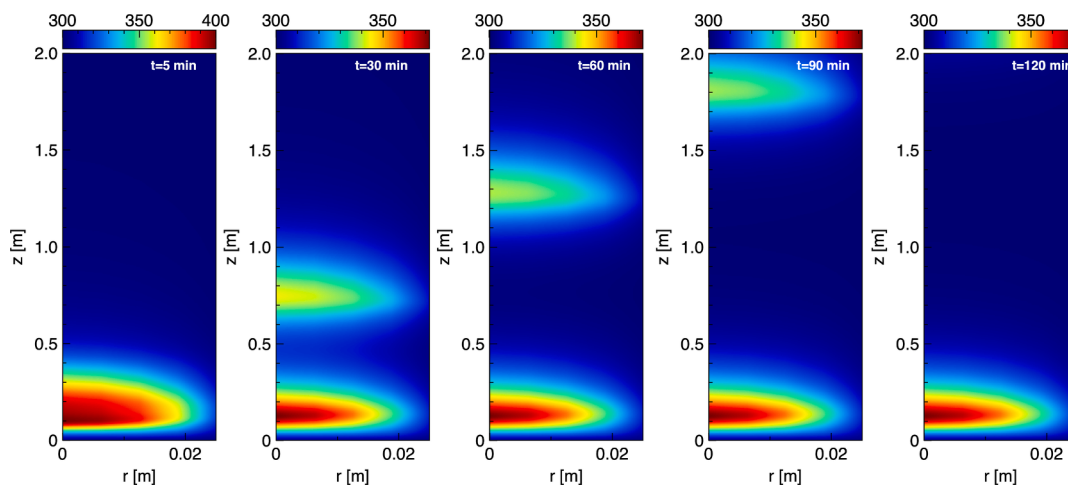


Fig. 3. Temperature ($^{\circ}\text{C}$) distribution at several time instants.

sorption enhanced conditions. The CO_2 conversion is thermodynamically limited by the high temperatures attained, thus, in the initial sections of the reactor, the temperature starts to decrease. Consequently, the hot region shrinks and anchors close to the reactor inlet (Fig. 3, $t > 5$ min). Moreover, in Fig. 3, for times greater than 5 min, a second temperature peak can be observed which progressively shifts towards the exit of the reactor. This is essentially due to the further conversion of the

reactants which are not consumed at the reactor inlet, that encounter downstream conditions in which SEM can still occur.

Fig. 4 shows the distribution of adsorbed water [$\text{mol}_{\text{H}_2\text{O}}/\text{kg}_{\text{ads}}$] for several times. The higher the temperature (see Fig. 3), the lower the amount of adsorbed water. Therefore, the distribution of adsorbed water (Fig. 4) mirrors that of temperature. For heights above 0.5 m, the effect of external cooling is predominant, and the temperature is almost

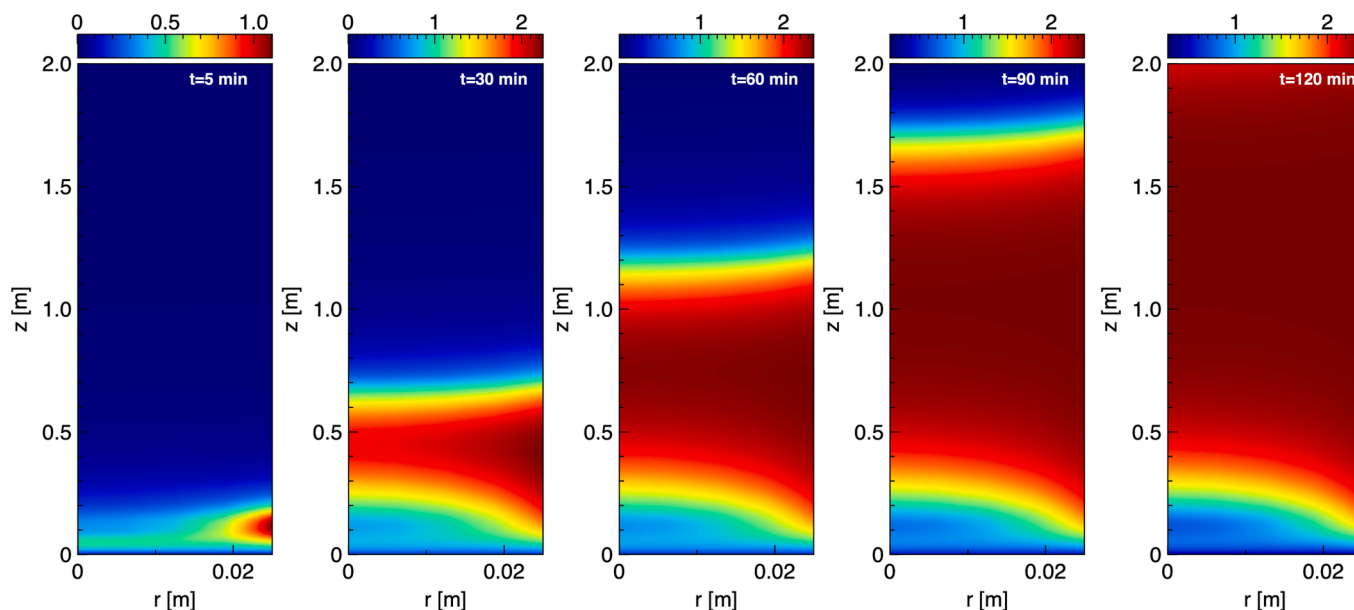


Fig. 4. Adsorbed water [$\text{mol}_{\text{H}_2\text{O}}/\text{kg}_{\text{ads}}$] distribution at several times.

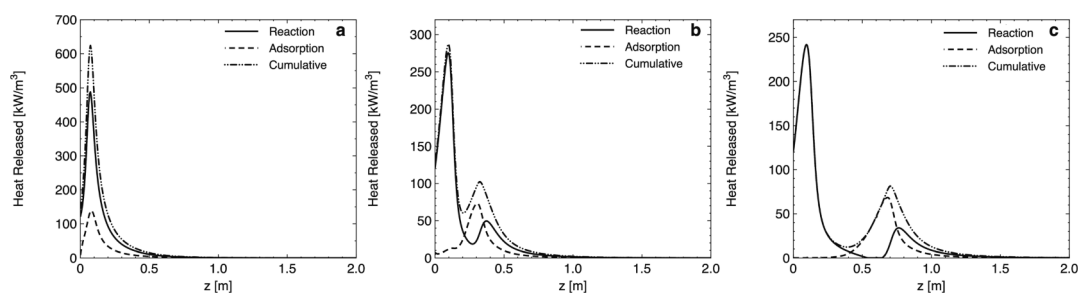


Fig. 5. Reaction, adsorption, and cumulative spatial profiles of the released heat at (a) $t = 5$ min, (b) $t = 30$ min, and (c) $t = 60$ min.

uniform. Consequently, the amount of water adsorbed onto the solid is also constant. On the contrary, the catalyst layers close to the inlet feature the least amounts of adsorbed water.

Fig. 5 shows the contribution to the total heat release of both the catalytic reaction and water adsorption at different times. At the beginning of the process, methanation and water removal simultaneously occur on the catalyst layers close to the inlet, so a maximum in the released heat spatial profile is observed (Fig. 5(a)). It can be noted that the heat released by methanation is approximately four times greater than that produced by adsorption, in agreement with data shown in Table 3. As the process goes on, the heat produced by methanation decreases and that of adsorption disappears on the first catalyst layers. At the same time, an additional methanation peak traveling along the reactor can be observed (Fig. 5(b)-(c)), slightly preceded by an adsorption peak. The adsorption of water makes indeed possible the further conversion of CO_2 into CH_4 . It is noteworthy that the heat released by adsorption is constant throughout the reactor, and in the innermost sections is higher than that produced by second methanation step. Indeed, in the innermost sections of the reactor, all the water produced upstream is adsorbed, while only unreacted CO_2/H_2 are transformed.

To better understand how the methanation reaction and the adsorption process occur through the reactor, and their consequences on system performance, Fig. 6(a-d) show the cross-sectional average values of: (a) – methanation reaction rate; (b) – adsorption rate; (c) – temperature; (d) – water adsorbed, while Fig. 6(e-h) report the cross-sectional average values of gaseous species molar fractions, at different reactor heights. To get insights on what happens in the catalyst layers closest to the reactor inlet, the cross-sections are not equally spaced, and the distance from the reactor inlet is reported in the legend.

It is recalled here that an increase in temperature boosts both the methanation and the adsorption rates. On the contrary, as the temperature increases, the maximum capacity of the solid to adsorb water decreases, and correspondingly decrease the breakthrough time, the positive effect of in-situ water removal, and ultimately the methane production. On the catalyst layers close to the inlet, the methanation reaction rate quickly increases, peaks, and then decreases (Fig. 6(a)). It is noteworthy that methanation velocity starts to decrease before the solid has reached its maximum adsorption capacity (Fig. 6(d)). Moreover, being the catalyst layers close to the inlet continuously fed by fresh reactants, methanation velocity is never zero in this region (dash-dot line in Fig. 6(a)). As suggested by Kiefer *et al.* (Kiefer *et al.*, 2022), the catalyst layers close to the inlet section must be regarded separately. Even if the solid is saturated by water in the external solid layer, methanation still occurs, producing water that will be adsorbed in the innermost sections, where conditions are such to promote SEM.

Therefore, the innermost sections provide further conversion of unreacted CO_2 and H_2 . In these sections, methanation only takes place under SEM conditions: it increases, reaches a maximum, and then decreases to zero (Fig. 6(a-b)), with almost similar peak values. For the adsorption rate, like for the methanation one, the highest peak is reached near the first catalyst sections. Indeed, on the catalyst layers close to the inlet, methanation and adsorption rates are the highest and the steeper, and the two processes are practically concurrent (Fig. 6(a-b)). As discussed above, the unconverted reactants are consumed downstream, but as the partial pressures are lower, the maximum reactions rate are lower and reaction fronts blurred. This also leads to lower temperatures (Fig. 6(c)). As the temperature decreases, the amount of water that can be removed increases (Fig. 6(d)), but the rate of the process decreases. Therefore, moving down the reactor, methanation and water removal become staggered. Moreover, at lower temperatures the reaction rate becomes limiting. Conversion becomes slower than convective mass transfer and slip of reactants occurs. This phenomenon is due to the limiting effect of the chemical reaction resulting in a lower conversion rate than convective mass transport (Kiefer *et al.*, 2022).

What has been discussed so far affects the gas composition at the reactor exit, and then the length of time during which the reactor produces pure methane. Besides a complete conversion, no trace of water is detected in the outgoing flow. After 70 min, the output methane molar fraction starts to decrease due to the slip of unreacted H_2 and CO_2 , while H_2O continues to be adsorbed in the last sections of the reactor where the solid has not yet reached saturation (Fig. 6(e-h)). The water breakthrough only occurs at around 88 min, and subsequently the percentage of methane starts to decrease quickly. Methane molar fraction decreases from $\sim 100\%$ down to 98% between 70 and 88 min, and then to about 88% between 88 and 120 min. Therefore, depending on specific regulatory limits for methane grid injection, the process should be arrested either at 70 min, to obtain a pure methane stream, or at 88 min, if 2% of H_2/CO_2 impurities are allowed. In the former case, the maximum working time of the reactor does not coincide with the breakthrough of water, and this results in a loss of approximately 20% of useful process time.

The limiting effect of the chemical reaction was also discussed in Kiefer *et al.* (Kiefer *et al.*, 2022), even experimentally for an isothermal reactor. The section-averaged values of molar fractions and CO_2 conversion at the reactor exit are reported in Fig. 7 for three different inlet temperatures according to the legend. In particular, the temperature at the reactor inlet was decreased/increased by 20°C . It is worth remembering that, in all the case investigated, initial temperature always coincided with the inlet and cooling one. As it can be seen in Fig. 7, an

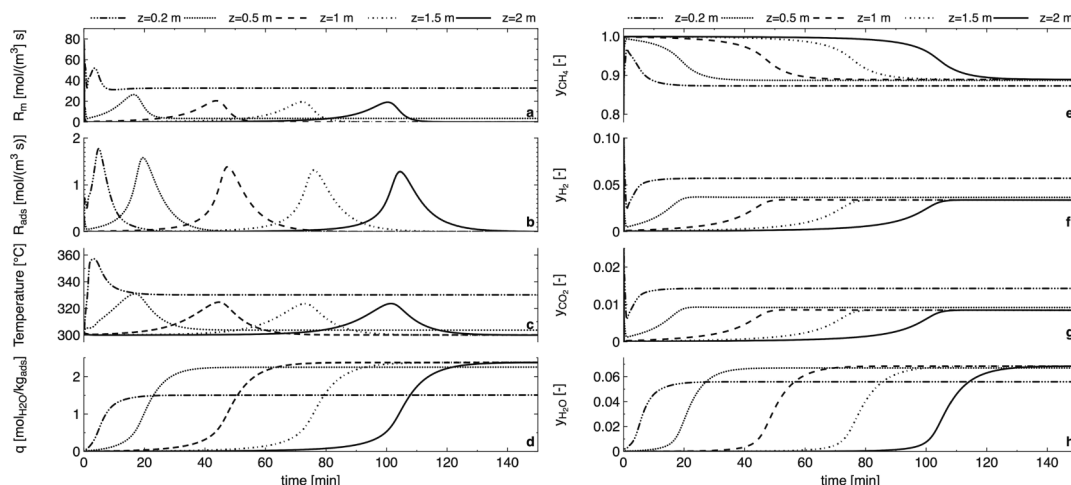


Fig. 6. (a) Average methanation reaction rate; (b) adsorption rate; (c) temperature; (d) adsorbed water; and (e-h) gas molar fractions.

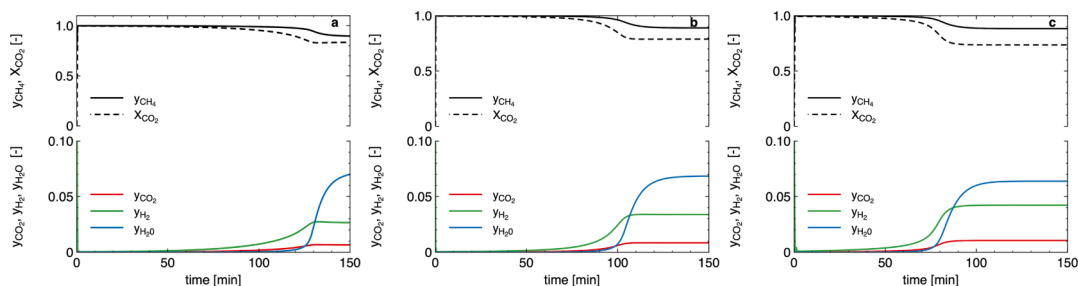


Fig. 7. Gas composition and conversion at reactor outlet: (a) $T = 280\text{ }^{\circ}\text{C}$; (b) $T = 300\text{ }^{\circ}\text{C}$; and (c) $T = 320\text{ }^{\circ}\text{C}$.

Table 4

Breakthrough time for H_2 and H_2O and their relative difference at different operating and inlet temperatures for an isothermal and cooled reactor.

Isothermal			
$T\text{ }[^{\circ}\text{C}]$	$t_{\text{bt,H}_2}\text{ [min]}$	$t_{\text{bt,H}_2\text{O}}\text{ [min]}$	$\Delta t_{\text{bt}}\text{ [%]}$
280	104.0	140.4	26
300	83.3	108.8	23
320	60.6	75.5	20
Non isothermal cooled reactor			
$T_{\text{in}}\text{ }[^{\circ}\text{C}]$	$t_{\text{bt,H}_2}\text{ [min]}$	$t_{\text{bt,H}_2\text{O}}\text{ [min]}$	$\Delta t_{\text{bt}}\text{ [%]}$
280	90.5	126.6	28
300	69.5	88.3	22
320	58.5	71.2	18

increase in temperature causes reactants slip to start closer and closer to the water breakthrough (i.e., the relative difference between $t_{\text{bt,H}_2\text{O}}$ and $t_{\text{bt,H}_2}$ decreases). Quantitative results were summarized in Table 4 for both the cooled packed bed reactor analysed in the present work, and for simulations performed with an isothermal packed bed reactor. Temperature plays a concurrent role. Indeed, increasing temperature rises reaction rates but the maximum adsorption capacity, and consequently the $t_{\text{bt,H}_2\text{O}}$, decreases.

Definitively, the limiting effect of the kinetics appears in a slip of the reactants, whose breakthrough always precedes that of water, with a more pronounced effect at lower temperatures, as already observed in Fig. 7. Moreover, a comparison between isothermal (see Supplementary Material S.5) and non-isothermal results suggests that isothermal model may predict longer breakthrough times than real one, overestimating the performance of the system. This result confirms the importance that the model developed in this work may have in supporting experimental activities and in providing reliable estimation of reactors performances for subsequent scale up.

4.2. Internal and external transfer limitations

In this Section, the mass and heat transfer limitations between the gas and solid phase, and within the catalyst pellet, were discussed. As previously addressed in Section 2.2, because of the solid conductivity being very high and the internal heat transfer resistance being negligible, only gas–solid heat transfer resistance was considered. The axial and radial temperature profiles for the two phases are quite similar. In all the investigated cases (not reported for the sake of brevity), the temperature of the solid is, at most, 2–5 $^{\circ}\text{C}$ higher than that of the gas.

Fig. 8 depicts the mole fractions of the four chemical species in the gas phase, on the surface of the solid, and in the centre of the solid, for the section close to the reactor inlet ($z = 0.2\text{ m}$).

In all the cases investigated, external mass transport resistances are less important than internal ones, and the maximum relative difference observed in the mole fraction of gas and surface is 7%. The effective diffusivity in the solid accounts for both molecular and Knudsen diffusivity. The values of the estimated diffusion coefficients have the same order of magnitude, so both Fick and Knudsen diffusion occur (the

Knudsen number in the pore catalyst is slightly greater than 1, meaning that the pore diameter and mean free path are similar). Thus, larger differences between surface and centre concentrations occur for CO_2 . Moreover, due to different diffusion limitations, the concentrations of the reactants differ from stoichiometric ones along the catalyst radius, particularly close to the central rod where the highest temperature is reached, and so the highest reaction velocity is attained. In each case, the concentration of the reactants is never zero at the center, which indicates that the entire catalyst is used in the SEM process. It is interesting to observe that, although water is a product of the methanation reaction, in the initial instants its concentration within the catalyst is lower than that at the surface, as in the case of H_2 and CO_2 . Indeed, at initial times, the water produced by the reaction is adsorbed within the solid until saturation, thus behaving as a reactant (Fig. 8). Then, when in a certain section the solid reaches saturation, water stops being removed from the gas phase, and the concentration within the catalyst particles increases. This is clearly shown in Fig. 9, that reports the H_2O molar fraction at longer times, for the same transversal section of Fig. 8.

Fig. 9 also highlights that the sections closest to the cooled wall were the firsts to attain maximum adsorption capacity. Indeed, when complete saturation is reached near the cooled wall, the lower temperatures allow for a greater conversion of the reactants, and this results in greater concentration gradients. A similar behaviour occurs in the innermost sections, on which, for longer times, methanation occurs in SEM condition. The diffusion of the reaction products from the centre of the catalyst towards the surface is limited by diffusion, resulting in a higher concentration of methane within the catalyst.

Finally, the effectiveness factor is defined as (Levenspiel, 1999):

$$\eta = \frac{\int_0^{V_p} R_m(y_s, T_s) dV_p}{V_p R_m(y_g, T_g)} \quad (29)$$

Fig. 10 shows the effectiveness factor at two different times, namely when methanation reaction and water removal occurs simultaneously ($t = 5\text{ min}$) and when all the solid on the section became saturated ($t = 20\text{ min}$), i.e., when water removal no longer occurs. The effectiveness factor shows a non-monotonic behaviour with time. It is significantly lower than 1 at the beginning (i.e., when SEM occurs), with minimum values of about 0.7, while increases to values close to 1 as the in-situ water removal does not occur anymore. To explain this significant difference, the Weisz-Prater number was used to determine the effective influence of internal diffusion limitations (Weisz and Prater, 1954):

$$N_{\text{WP}} = \frac{R_m \rho_{\text{cat}} d_p^2}{4 D_{\text{eff},i} c_i} \quad (30)$$

If $N_{\text{WP}} < 1$, internal diffusion limitations can be neglected (Weisz and Prater, 1954). The N_{WP} calculated for the two reactant species decreases significantly with time. When the process takes place under SEM conditions, higher reaction rates are achieved due to the in-situ water removal, and the internal resistances to the transport of matter cannot be neglected. However, as the removal of water no longer occurs, diffusion limitations become negligible.

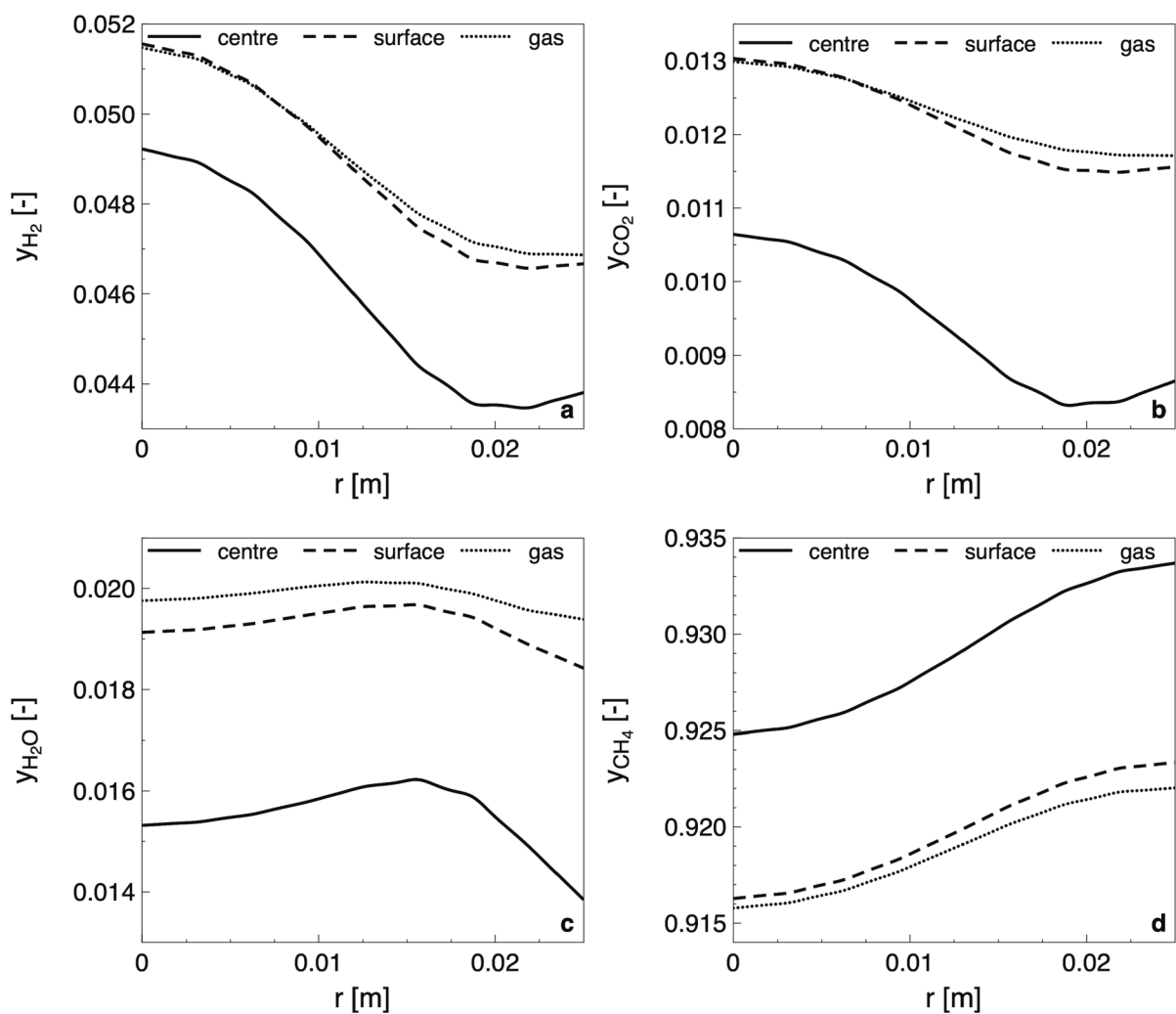


Fig. 8. Gas molar fraction profiles in the centre and on the surface of the solid, and in the gas, along a transversal section at 0.2 m from the reactor entrance for $t = 5$ min.

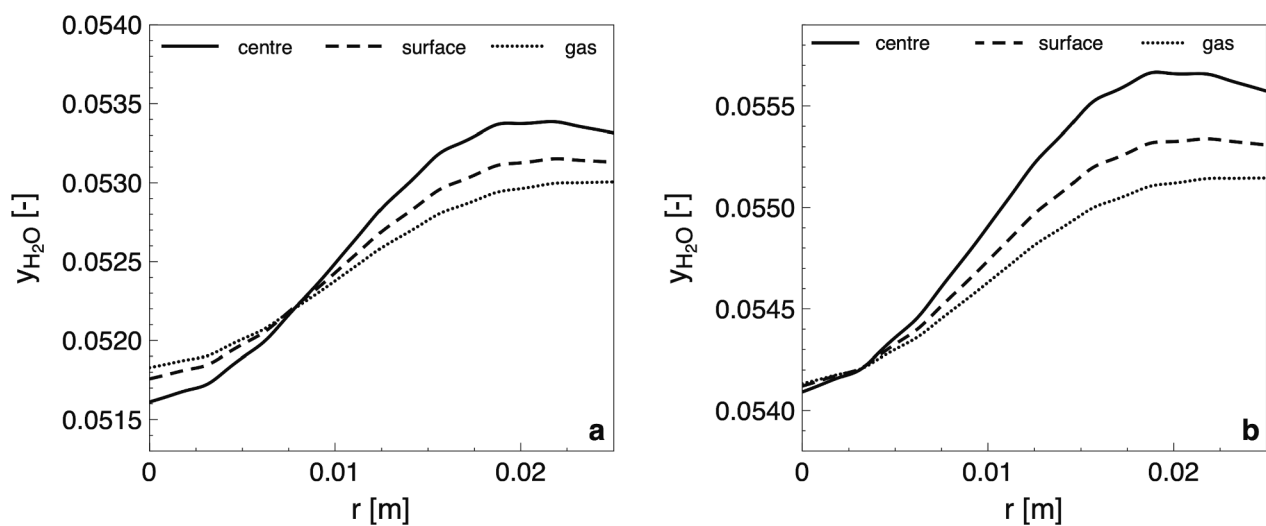


Fig. 9. H_2O molar fraction profiles in the pellet at the catalyst centre, surface, and in the gas, along a transversal section at $z = 0.2$ m; (a) $t = 10$ min; and (b) $t = 20$ min.

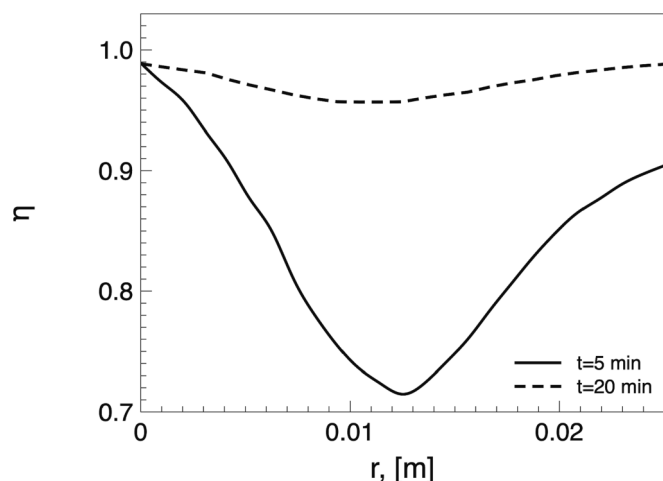


Fig. 10. Effectiveness factor profiled along the radial coordinate at two different times (transversal section $z = 0.2$ m).

4.3. Sensitivity analysis: Effect of GHSV and pressure on system behaviour

Gas space velocity is a key parameter for the system behaviour. Increasing the GHSV causes an increase in the flow rates but, on the other hand, reduce reactants residence time and thus, their conversion. Wei *et al.* (Wei *et al.*, 2021a) experimentally evaluated water breakthrough capacity in isothermal conditions for SEM, finding that increasing GHSV slightly affects the water breakthrough capacity that, mainly depends on the operating temperature. The simulations performed in this work using the non-isothermal cooled reactor led to wider conclusions, because the different temperature distributions obtained significantly affected the distribution of water adsorbed onto the solid during the process. Fig. 11 shows the temperature ($^{\circ}\text{C}$) and the adsorbed water [$\text{mol}_{\text{H}_2\text{O}}/\text{kg}_{\text{ads}}$] when water breakthrough occurred, for increasing values of GHSV. It is evident that as GHSV increases, the methanation reaction starts into the innermost layers of the reactor. The maximum temperatures reached do not change significantly for the different GHSV values considered, but the increase in convective heat transport spreads the heat produced, expanding the hot area inside the reactor. This directly affects the adsorbed water, which significantly decreases as GHSV increases.

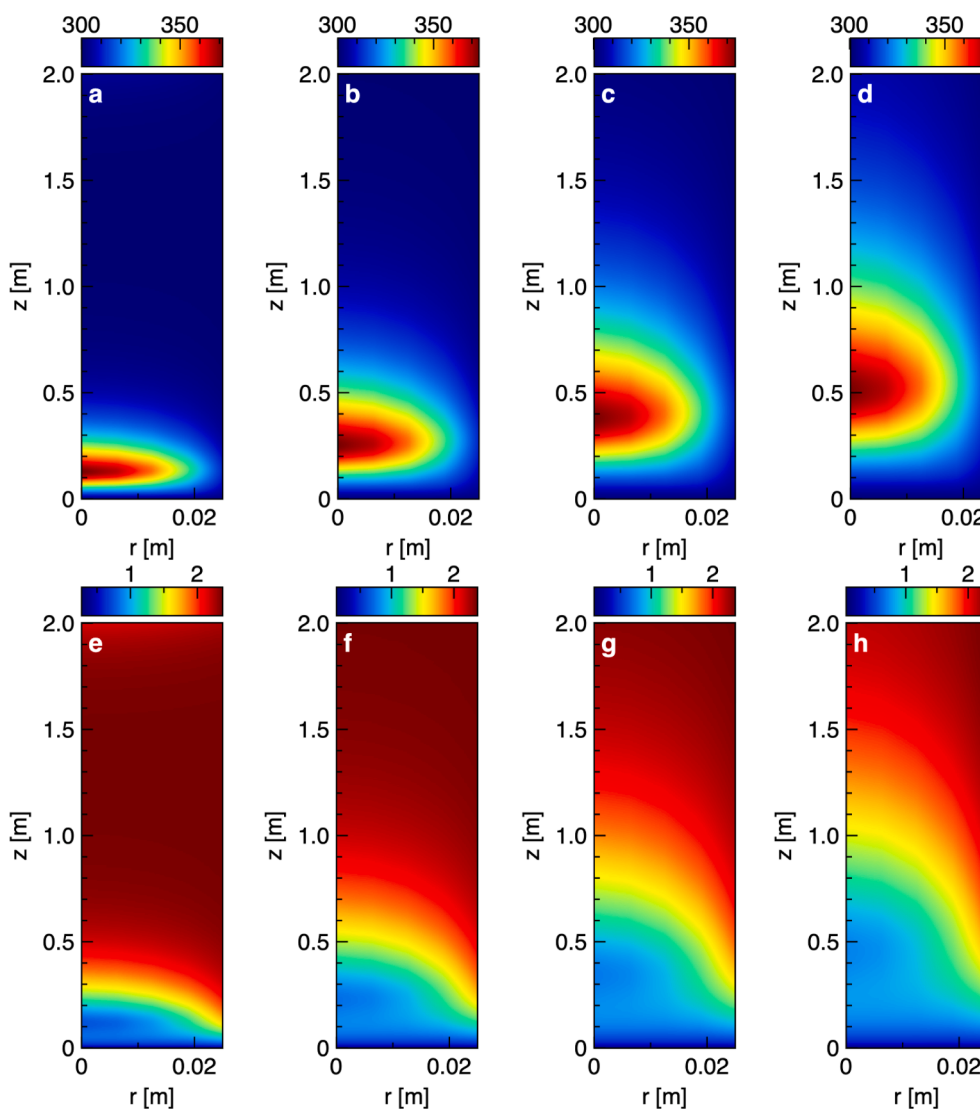


Fig. 11. (a-d): Temperature ($^{\circ}\text{C}$) and (d-h) adsorbed water ($\text{mol}_{\text{H}_2\text{O}}/\text{kg}_{\text{ads}}$) distribution at $t = 120$ min; (a) and (e) GHSV = 0.5; (b) and (f) GHSV = 1; (c) and (g) GHSV = 1.5; (d) and (h) GHSV = $2 \text{ m}^3 \cdot \text{kg}_{\text{ads}}^{-1} \cdot \text{h}^{-1}$.

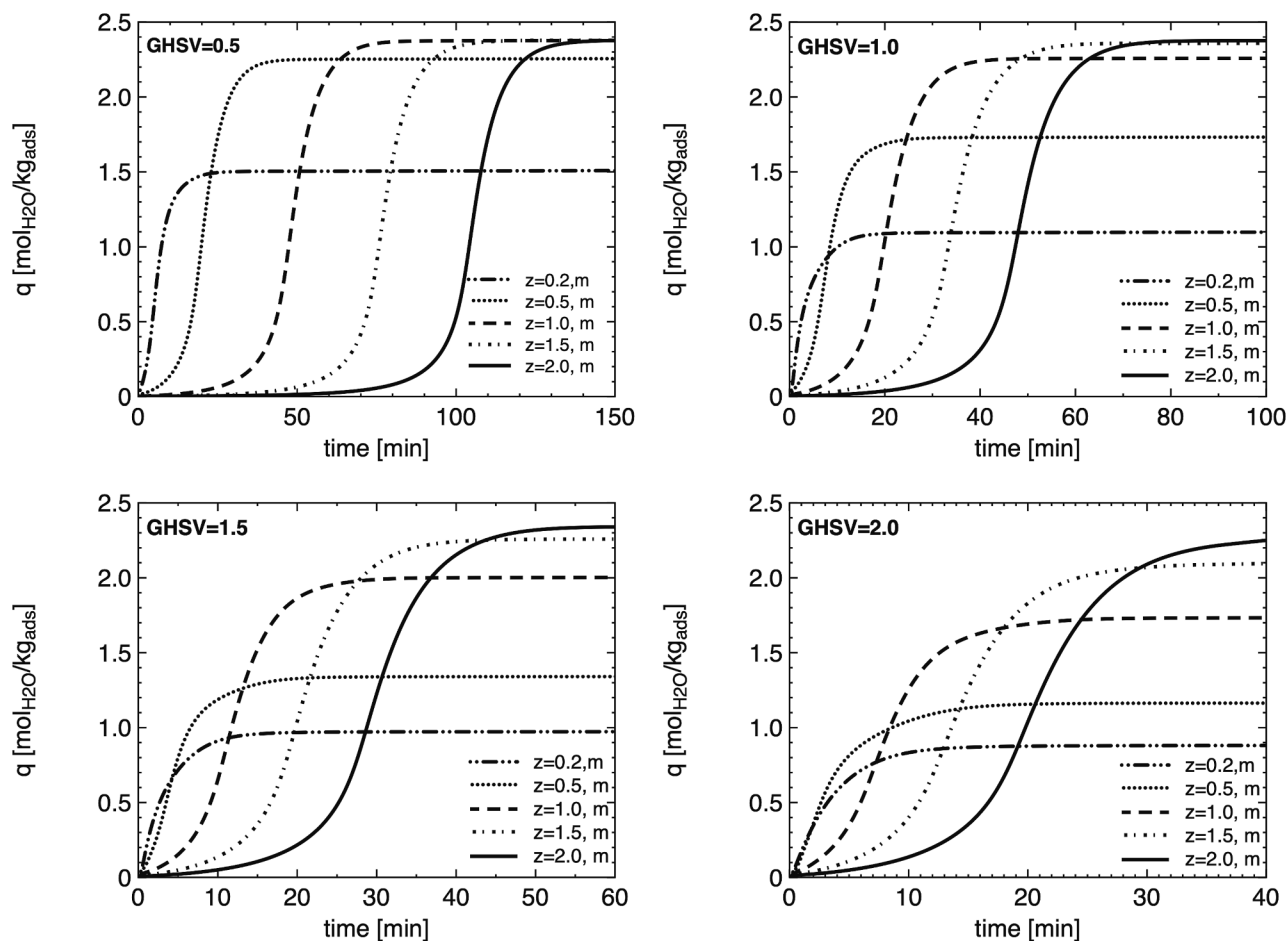


Fig. 12. Averaged adsorbed water on transversal section versus time at different reactor height for several GHSV ($\text{m}^3 \cdot \text{kg}_{\text{ads}}^{-1} \cdot \text{h}^{-1}$).

Table 5

Breakthrough time for H_2 and H_2O , their percentage difference, and overall adsorbed water for several GHSV values.

GHSV [$\text{m}^3 \cdot \text{kg}_{\text{ads}}^{-1} \cdot \text{h}^{-1}$]	$t_{\text{bt,H}_2}$ [min]	$t_{\text{bt,H}_2\text{O}}$ [min]	$\Delta t_{\text{bt}} [\%]$	$\text{mol}_{\text{H}_2\text{O}}/\text{kg}_{\text{ads}}$
0.5	69.5	88.3	22	2.48
1	22.4	42.1	47	2.39
1.5	10.6	23.3	56	1.98
2	5.6	14.0	60	1.58

A better insight can be obtained by analysing the average adsorbed water on transversal section at different reactor heights, depicted in Fig. 12 for all the investigated GHSV values. By increasing the space velocity of the gas, the adsorption of water in the innermost sections starts earlier in time, but the reached higher temperatures reduce the maximum amount of water that can be adsorbed. For example, on the section at $z = 0.5$ m, q decreases from 2.2 to 1.1 $\text{mol}_{\text{H}_2\text{O}}/\text{kg}_{\text{ads}}$ when GHSV is increased from 0.5 to 2 $\text{m}^3 \cdot \text{kg}_{\text{ads}}^{-1} \cdot \text{h}^{-1}$, respectively.

Finally, for all the GHSV values investigated, the moles of adsorbed H_2O per mass of solid were calculated when the solid reached saturation. The adsorbed water decreases from 2.48 to 1.58 $\text{mol}_{\text{H}_2\text{O}}/\text{kg}_{\text{ads}}$ when GHSV is increased from 0.5 to 2 $\text{m}^3 \cdot \text{kg}_{\text{ads}}^{-1} \cdot \text{h}^{-1}$, as detailed in Table 5. Therefore, accounting for heat transfer and generation as in present work, highlights that water breakthrough capacity can be significantly affected by the GHSV. This result further emphasizes the importance that the model developed in this work can have in predicting variation of system performance when changing the process parameters.

The analysis of Table 5 also reveals that increasing the GHSV leads to rising reductions of $t_{\text{bt,H}_2\text{O}}$: doubling GHSV from its lower value almost halves $t_{\text{bt,H}_2\text{O}}$, but quadrupling it results in a reduction by 6 of the water breakthrough time. Furthermore, it should be recalled that due to the reactants slip, the time interval during which the system produces pure methane is lower than $t_{\text{bt,H}_2\text{O}}$ and mostly equal to $t_{\text{bt,H}_2}$. As GHSV increases, the transport by convection is faster and faster and this brings to an even stronger reduction of $t_{\text{bt,H}_2}$, as shown both in Table 5 and Fig. 13. The present results partly contrast with what reported in literature by some Authors (Borgschulte et al., 2013; Walspurger et al., 2014), who stated that in isothermal conditions a complete CO_2 conversion can be obtained for quite long times even at high GHSV values. For example, for a $\text{GHSV} = 2.5 \text{ m}^3 \cdot \text{kg}_{\text{ads}}^{-1} \cdot \text{h}^{-1}$, as in Walspurger et al. (Walspurger et al., 2014) in the case of non-isothermal cooled reactor, the water breakthrough occurs after approximately 8 min but the CO_2 conversion is always lower than 0.94 due to slip of reactants.

The results discussed so far refer to a pressure value of 1 bar. As the pressure increases, the gas velocity decreases as well as GHSV. Moreover, the pressure also has a positive effect on conversion. Indeed, CO_2 methanation is usually carried out at pressures greater than 10 bar (Götz et al., 2016), but selective adsorption via membrane reactors may be more convenient in such cases (Massa et al., 2020). To pursue process intensification, the effect of two different pressure values has been considered in this work. The main results of this analysis are summarized in Table 6 and Table 7 for $P = 3$ and 5 bar, respectively.

As reported in Table 6 and 7, the increase in pressure leads to a significant increase in both reactants and water breakthrough times.

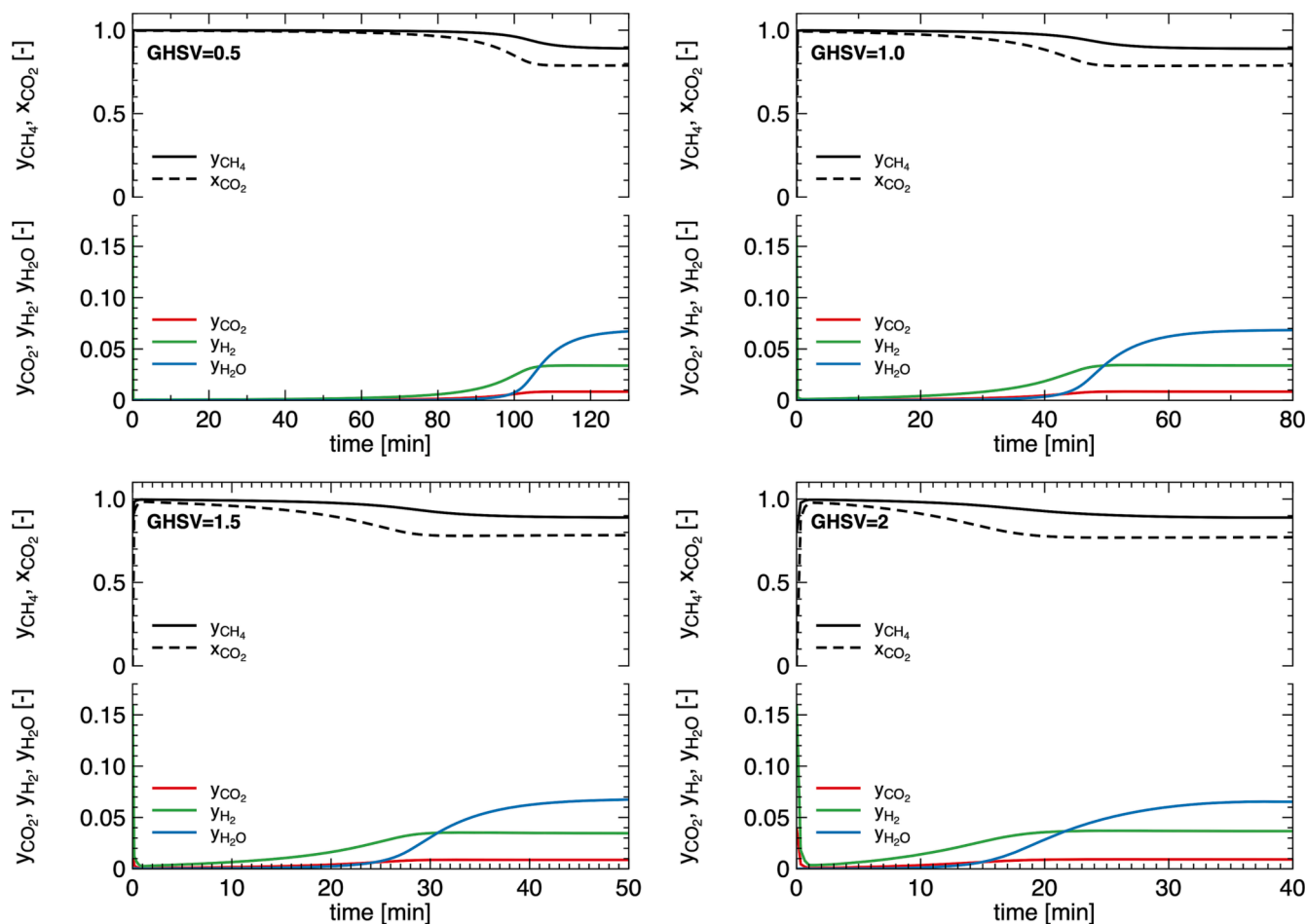


Fig. 13. Gas composition and conversion at reactor outlet for several GHSV values ($\text{m}^3 \bullet \text{kg}_{\text{ads}}^{-1} \bullet \text{h}^{-1}$).

According to Eq.(23), the maximum adsorption capacity (q^*) increases for higher pressure. Moreover, as the pressure increases, the velocity decreases leading to a contraction of the hot area in the reactor, so the overall adsorbed water increases. For example, at $\text{GHSV} = 2 \text{ m}^3 \bullet \text{kg}_{\text{ads}}^{-1} \bullet \text{h}^{-1}$, the overall adsorbed water for mass of solid increases from 1.58 $\text{mol}_{\text{H}_2\text{O}}/\text{kg}_{\text{ads}}$ at 1 bar (see Table 5) to 3.74 and 4.76 $\text{mol}_{\text{H}_2\text{O}}/\text{kg}_{\text{ads}}$ at $P = 3$ and 5 bar respectively. These two effects explain the increased time of $t_{\text{bt,H}_2\text{O}}$.

In addition, the increase in pressure has a beneficial effect on reaction velocity by reducing the slip of reactants. As it can be seen from the results shown in Tables 5-7, when the pressure increases the relative difference between the breakthrough times decreases. At the same time, it must be considered that at higher pressures the maximum temperatures reached during the process also increase. As discussed in Section

4.1, the highest temperatures occur at the beginning of the process, in sections near the reactor inlet and along central rod. Fig. 14 reports the temperature profiles along the central rod at the initial instants (a) and when the water breakthrough occurred (b).

In all the cases investigated, the maximum temperature reached is below 550°C , that is the threshold temperature at which Ni sintering can occur (Rönsch et al., 2016). At the same time, it is important to note that at higher temperature values, the phenomenon of carbon formation could be relevant, especially due to the decrease in process yield. Only a more detailed description of the process including the solid carbon formation reactions and a technical-economic analysis will make it possible to define the best operating conditions.

Table 6

Breakthrough time for H_2 and H_2O , and their percentage difference for several GHSV values at $P = 3$ bar.

GHSV [$\text{m}^3 \bullet \text{kg}_{\text{ads}}^{-1} \bullet \text{h}^{-1}$]	$t_{\text{bt,H}_2}$ [min]	$t_{\text{bt,H}_2\text{O}}$ [min]	$\Delta t_{\text{bt}}[\%]$
0.5	152.7	171.3	10
1	62.8	77.5	19
1.5	35.6	47.8	25
2	21.3	33.3	36
2.5	13.3	23.8	43
3.0	9.6	18.5	48

Table 7

Breakthrough time for H_2 and H_2O , and their percentage difference, for several GHSV values at $P = 5$ bar.

GHSV [$\text{m}^3 \bullet \text{kg}_{\text{ads}}^{-1} \bullet \text{h}^{-1}$]	$t_{\text{bt,H}_2}$ [min]	$t_{\text{bt,H}_2\text{O}}$ [min]	$\Delta t_{\text{bt}}[\%]$
0.5	199.8	212.5	6
1	85.7	99.8	14
1.5	50.5	61.1	17
2	33	42.5	22
2.5	22.3	31.8	30
3.0	16.1	25.5	37

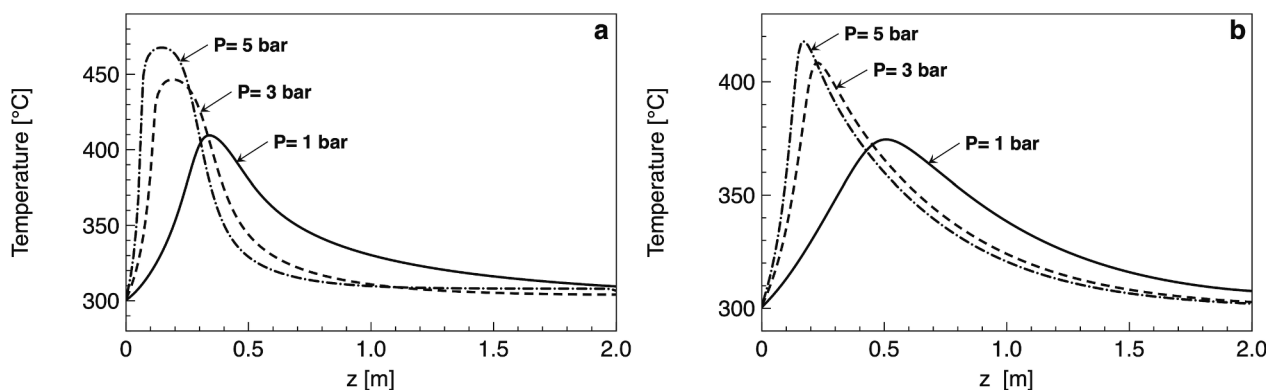


Fig. 14. Temperature profiles along the central rod for $GHSV = 2 \text{ m}^3 \cdot \text{kg}_{\text{ads}}^{-1} \cdot \text{h}^{-1}$, for several pressure values.

5. Conclusions

In this work the reaction/adsorption step of a sorption-enhanced methanation process occurring in a cooled packed-bed reactor is modelled by a detailed 2-dimensional, heterogeneous, and dynamical model. The reactor is considered to be filled by a Ni-loaded 13X zeolite where the Ni acts as catalyst and zeolite promotes in-situ water removal.

The mathematical model was validated through experimental data available in literature and a very good agreement in terms of water adsorption capacity, CO_2 conversion, and breakthrough times of water and reactants was found.

Obtained numerical results show that:

- the heat distribution and consequently the temperatures inside the reactor are not uniform. Gradients of about 100°C exist along both the reactor axis and radius;
- temperature maximum is reached on catalyst layers close to the inlet section at the very beginning of the process. Under all the investigated operating conditions, it never exceeds 470°C ;
- the distribution of adsorbed water mirrors that of temperature, thus only a detailed description of the system allows an accurate prediction of the water breakthrough time;
- reaction rate is limiting, so decreasing the inlet temperature makes the conversion slower than convective mass transfer and slip of reactants occur. This phenomenon results in reactants breakthrough time lower than water one;
- a pure SNG flow can be produced until reactants breakthrough occur. In the time range between water and reactants breakthrough, methane molar fraction in the leaving current is still greater than 98%;
- external and internal mass transfer limitations mainly take place on catalyst layers where methanation occurs under SEM condition;
- for any given operating pressure, as GHSV water increases breakthrough times decrease non-monotonically;
- processes intensification can be pursued by a moderate increase in operating pressure with respect to atmospheric one. Indeed, for typical GHSV values reported in the literature ($2\text{--}2.5 \text{ m}^3 \cdot \text{kg}_{\text{ads}}^{-1} \cdot \text{h}^{-1}$), water breakthrough times exceed 23 min at 3 bar and reach 32 min at 5 bar.

Ultimately, it can be inferred that the knowledge of reactor heat distribution is of utmost importance to properly design a SEM process. Due to the contrasting effects of temperature on both the reaction/adsorption rates and the water sorption capacity, it is mandatory to accurately evaluate the system breakthrough times and how they are affected by the process parameters to achieve very high conversion degrees and/or to avoid reactants slip. Moreover, it should be considered that, in the design of a commercial unit, several reactors arranged in parallel configuration would be required to achieve a continuous

methane production. Indeed, while in one or more reactors SEM occurs, several others need to be regenerated through sorbent drying. Prediction of the breakthrough times is a fundamental prerequisite to optimize the performance of the system in terms of, for example, the number of reactors to be operated in parallel and their synchronization across the different process steps (e.g., adsorption/reaction, purge, regeneration), also in the perspective of a process intensification. Furthermore, the optimal choice of operating conditions allows for quite long time intervals during which high-purity methane is attained, even at low pressure values, opening promising perspectives for this technology.

CRediT authorship contribution statement

Piero Bareschino: Validation, Visualization, Writing – original draft, Writing – review & editing. **Giuseppe Piso:** Writing – review & editing. **Francesco Pepe:** Funding acquisition. **Claudio Tregambi:** Validation, Visualization, Writing – original draft, Writing – review & editing. **Erasmus Mancusi:** Conceptualization, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ces.2023.118876>.

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